

Data response and quantitative skills

Content

1. Using the extract

Data response questions test whether you can use the evidence you are given. The extract is not background reading; it is part of the answer.

Read the question first. Note the command word, the marks and what to look for in the extract.

Skim the extract for evidence. Highlight figures, dates, names of firms and any cause-and-effect statements.

Choose evidence that supports a point. Do not quote for the sake of it. Use a fact at the step in your chain where it fits.

Quote briefly and interpret. "Sales fell from 2.4 million to 1.9 million (Figure 1), a fall of 21%, so revenue fell unless price rose." Then say what it means for the argument.

Use it to evaluate. Extracts often hint at short run versus long run, who is affected, or a constraint that limits an effect.

Avoid

Copying long sentences from the extract. Marks go for applying the evidence, not for repeating it. Do not use knowledge that contradicts the data given.

2. Calculations: formulas and layout

Show the formula, the substitution and the answer with units or %. Follow any rounding instruction in the question. If the working is right but the final answer is wrong, you can still earn marks.

Measure	Formula
Percentage change	$(\text{new} - \text{old}) \div \text{old} \times 100$

Price elasticity of demand	$\% \text{ change in quantity demanded} \div \% \text{ change in price}$
Total revenue	$\text{price} \times \text{quantity}$
Average cost	$\text{total cost} \div \text{quantity}$
Profit	$\text{total revenue} - \text{total cost}$
Profit margin	$\text{profit} \div \text{total revenue} \times 100$
Concentration ratio (n-firm)	sum of the market shares of the n largest firms
Market share	$\text{firm's sales} \div \text{total market sales} \times 100$

Watch for

- **Percent vs percentage points.** A rate rising from 4% to 6% is a rise of 2 percentage points, and a 50% rise.
- **Sign of PED.** Price and quantity move in opposite directions, so PED is negative. The size (ignoring the sign) tells you whether demand is elastic or inelastic; check what the question asks for.
- **Use the original value** as the base for percentage change.
- **Units.** £, millions, thousands, per cent. Copy them from the table.

3. Interpreting graphs and tables

1. **Read the title, axes and units first.** Are the values in £, %, millions, an index?
2. **Describe the trend** in a sentence: direction, size, the time period, any turning points.
3. **Put numbers on it.** "Rose from 2.1 to 3.4" beats "rose a lot".
4. **Compare and calculate** where useful: percentage change, share of the total, difference between series.
5. **Explain why** using economics, not just what. Then say what it means for the question.
6. **Check the source and date.** A short time series may not show the long-run picture.